Exploring Novelty of Suno AI Lyrics

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***Abstract*—We aim to investigate how pop lyrics generated from the popular Suno AI music generation tool differ semantically from human-written pop songs. Using cosine distances between SBERT sentence embeddings, we define novelty and examine related metrics such as variance and line-to-line angles [1]. We find clear differences in centroid-level novelty between Suno and human lyrics and smaller, though measurable, differences in related metrics.**

I. Motivation

Suno AI, a popular generative AI music tool, recently entered into a $500 million partnership with Warner Music Group. Suno, formerly a controversial company within the music industry, is now positioned alongside one of the largest organizations in the field. This motivates investigation into how this will affect musical products in the pop music industry. Few statistical analyses have been done on the outputs of music generation platforms like Suno and Udio. We aim to gain insight into the semantic space of Suno pop lyric outputs.

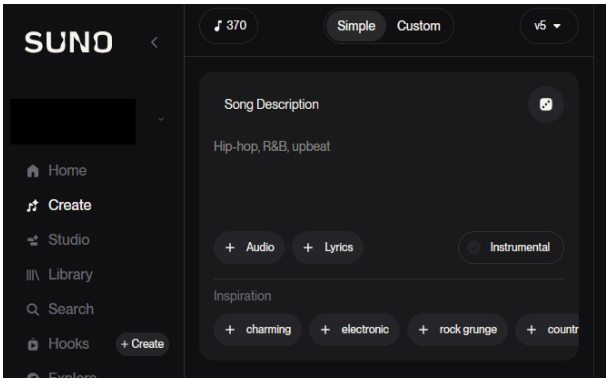


Fig. 1: Suno AI prompt interface.

II. Introduction

Suno AI is capable of generating human-like songs in popular genres from user prompts and various adjustable parameters. With Suno and tools like it producing complete tracks with vocals and lyric texts, lyric-level analysis provides an accessible way to examine properties of AI-generated music. We investigate the semantic embedding space of Suno-generated and human-written pop lyrics focusing on novelty as measured by cosine distances between line embeddings, song centroids, and group centroids.

Many statistical traits of AI-generated media have been uncovered. Mitchell et al. use perturbation-based token probability methods to classify AI and human-written text. [2]. Zaitsu et al. leverage stylometric features with random forest classifiers [3]. Tulchinskii et al. analyze intrinsic dimensionality of embeddings [4]. These approaches show that semantic and stylistic characteristics can separate AI-generated from human-generated text, suggesting that lyric embeddings of Suno outputs may reveal model-specific tendencies.

III. Related Work

A. AI-generated music and text detection

Detection of AI musical content is challenged by audio noise and format variability, but text classification—when available—is often more robust. [2, 4]. Frohmann et al. transcribed sung lyrics using ASR, computed embeddings, and trained an MLP to classify AI vs. human lyrics [5]. This method outperformed audio-based benchmarks. These works demonstrate that semantic embeddings of lyrics can show differences in vector space and support reliable classification of AI and human origins.

B. Semantic Novelty Measures

Prior work motivates embedding-based analysis for distinguishing AI and human text. Sentence embeddings are widely used to assess semantic similarity. Akinwande et al. used SBERT embeddings to classify AI text [6]. Casini et al. used embedding visualizations to study Suno and Udio platforms and user behaviors [7]. This shows that embedding-based analyses of Suno lyrics has empirical grounding.

Cosine distance, a common measure of novelty and semantic similarity in embedding analysis, measures the relatedness of two text passages. It correlates with perceived creativity in problem-solving contexts [8] and scientific novelty judgments [9]. These results motivate applying cosine-distance-based novelty measures to Suno-generated lyrics.

IV. Method

A. Inputs and Flowchart

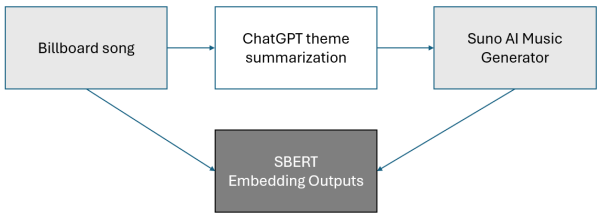


Fig. 2: Flowchart of the system.

TABLE I: Comparison of Song-Level Novelty Metrics (Paired t-Test. n=280 AI, n=280 Human)

Metric	Suno	Human	t-stat	p-value
line to song centroid distance mean	0.653 726	0.659 160	1.00	0.318 011
song centroids to group centroid distance mean	0.586 530	0.831 602	16.17	0.000 000
song centroids to global centroid distance mean	0.596 897	1.141 270	43.70	0.000 000
line to song centroid distance variance	0.018 093	0.041 456	13.66	0.000 000
song centroids to group centroid distance variance	0.012 547	0.010 067	−8.12	0.000 000
song centroids to global centroid distance variance	0.013 048	0.010 343	−7.73	0.000 000

Figure 2 shows our symmetrical data-gathering pipeline. We collected 284 Billboard Top 100 songs tagged as "pop" in MusicBrainz and generated theme-matched Suno pop lyrics using descriptions produced by ChatGPT [10]. Using SBERT (all-mpnet-base-v2), we computed embeddings for every line of each song. This dataset supports distance-based novelty analyses, clustering, dimensionality reduction, line-transition analysis, and classification experiments.

B. Novelty Evaluation Metrics

Novelty is defined as cosine distance between an embedding vector and a corpus centroid. We compute:

- mean cosine distance between song lines and the song centroid
- mean cosine distance between each song centroid and its group (Human or Suno) centroid
- mean cosine distance between each song centroid and the global centroid

Larger cosine distances correspond to greater semantic dissimilarity. We also compute clustering measures and line-to-line angle and distance metrics. For classification, we use the random forest classifier from Sci-Kit Learn along with its GroupKFold functionality to prevent data leakage. [11].

V. Design

A. Tests

We evaluate the corpus using k-means clustering; dimensionality reduction; cosine distances and their variances at both the song and line levels; angles and angle variances between successive lines; and random forest classifiers that use various combinations of these metrics as parameters. We use angle metrics in radians in our line transition analysis as an interpretable measure of line-to-line trajectory.

B. Methodology

1) Obtain embedding data for all lyric lines.
2) Perform clustering and t-SNE visualization.
3) Compute novelty metrics and paired statistical significance at the song level.
4) Compute line-to-line transition metrics and unpaired statistical significance.
5) Train classifiers with Sci-Kit Learn.

VI. Results

A. Song-Level Analysis

No meaningful clustering was found with k-means. Table I above summarizes our resulting novelty metrics. Human song centroids are farther from both group and global centroids than Suno centroids. Line-to-song centroid distances are the same across the two groups.

B. Line Transition Analysis

We restricted our inputs for the line transition analysis to songs with greater than 15 lines but less than 75 lines. Line-transition metrics show no significant differences across groups. These metrics are also useless in classification tasks. Line transition information can be found in in Table II in the Appendix. Classifier results can be found in Table III in the Appendix.

C. Classification Results

Using GroupKFold to separate matched pairs, a classifier trained on centroid metrics achieves 93% accuracy. A classifier using only line-transition metrics reaches 50%. Combining all metrics also reaches 93%. The classifier results are in Table III in the Appendix.

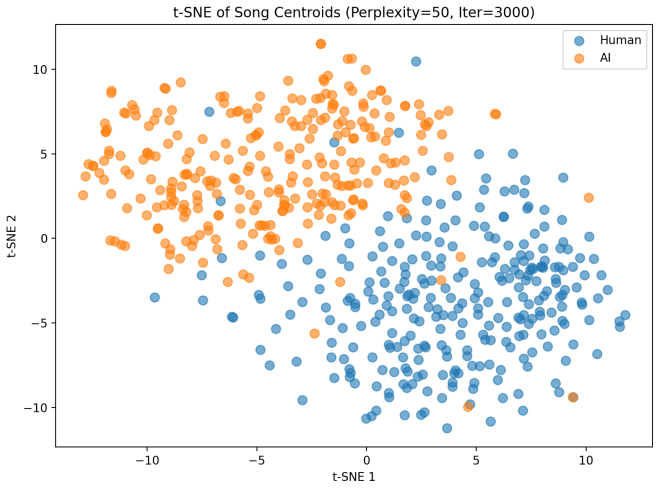


Fig. 3: t-SNE Dimension Reduction of Human (Blue) and Suno-Generated (Orange) Lyrics.

VII. Discussion

A. Limitations

- Suno parameters such as "weirdness" and other custom settings were not varied.
- Song structures often differed between human and Suno versions. This is evidenced by human songs being on average double the line count of AI-generated songs. Therefore, the only truly paired comparison is at the song centroid level.
- Human lyric collection was capped at 284 songs due to access constraints. Our dataset was limited to 568 total songs.
- Our corpus was intentionally homogenized to enable clear comparisons, but this design choice may limit generalization to more experimental genres.

B. Discussion

Human songs occupy a more dispersed region of embedding space than Suno outputs, reflected by larger distances from group and global centroids. This suggests average human song is more novel than its Suno counterpart. It also confirms the results of previous studies [5] [2] which find AI and human outputs clearly separable in vector space. Line transition metrics are not significantly different across Human and Suno groups and provide no predictive signal. These results point toward centroid-based metrics as the clearest examples of semantic patterns in Suno-generated pop lyrics.

We expected genre- and topic-specific structure in k-means clustering but observed no meaningful clusters, likely due to the homogeneity of the Billboard pop lyric dataset. In contrast, t-SNE embeddings of song centroids and simple classifier analyses reveal clear separation between groups. We also anticipated differences in line-transition metrics; Suno's ability to produce human-like semantic transitions may explain why its lyrics appear human-like yet remain separable under other measures. Further analysis of line-trajectory features is warranted.

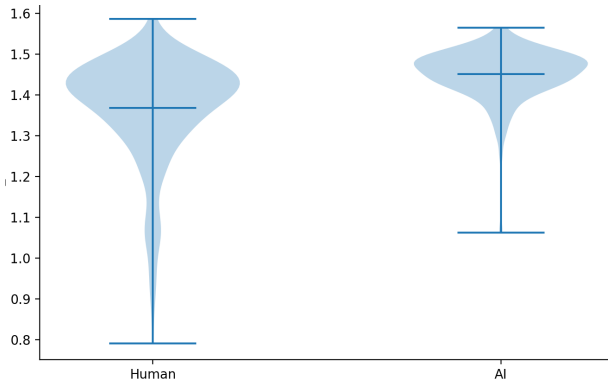


Fig. 4: Violin plot of mean angles between successive lines in radians.

We also attempted thematic analysis to assess how specific keywords influence novelty or line trajectories but were unable to obtain reliable results. This remains a direction for future work.

C. Contribution

Little work transparently characterizes the statistical structure of Suno AI outputs. Given Suno's growing adoption, it is important to evaluate how the system may alter user creative behavior. Future studies should analyze Suno outputs under varied generation parameters or through its integrated DAW to assess workflow-driven effects. Other systems, such as Udio, warrant similar examination.

Our results suggest that text-only prompting, with minimal human intervention, produces a relatively homogeneous output distribution. Related work in visual art reports a similar pattern: AI-assisted creation reduced average content novelty while increasing peak content novelty over time.[12] Examining Suno's full parameter space may reveal an analogous effect in music generation.

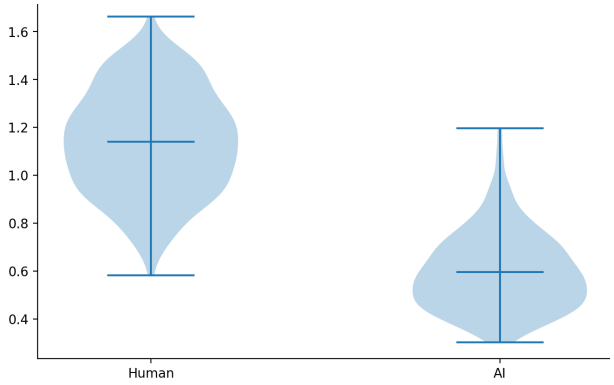


Fig. 5: Violin plot of song centroid distances from global centroid.

VIII. Conclusion

We examined the semantic embedding space of Suno-generated and human-written pop lyrics using centroid analyses, clustering, and line-level transition metrics. Across this range of methods, we evaluated the extent to which different embedding-derived features characterize the semantic patterns of AI-generated and human-written songs. We also assessed the usefulness of these features in classification tasks using random forest models informed by centroids, transition metrics, and combinations thereof. Embedding-based novelty metrics reveal clear centroid-level differences between Suno and human pop lyrics, while line-transition metrics provide no useful cues. These findings indicate stable semantic properties of Suno outputs and prompt deeper investigation into generative AI music tools.

REFERENCES

[1] N. Reimers and I. Gurevych, "Sentence-bert: Sentence embeddings using siamese bert-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 11 2019. [Online]. Available: https://arxiv.org/abs/1908.10084

[2] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, and C. Finn, "Detectgpt: Zero-shot machine-generated text detection using probability curvature," 2023. [Online]. Available: https://arxiv.org/abs/2301.11305

[3] W. Zaitsu, M. Jin, S. Ishihara, S. Tsuge, and M. Inaba, "Stylometry can reveal artificial intelligence authorship, but humans struggle: A comparison of human and seven large language models in japanese," *PLOS ONE*, vol. 20, no. 10, pp. 1–18, 10 2025. [Online]. Available: https://doi.org/10.1371/journal.pone.0335369

[4] E. Tulchinskii, K. Kuznetsov, L. Kushnareva, D. Cherniavskii, S. Barannikov, I. Piontkovskaya, S. Nikolenko, and E. Burnaev, "Intrinsic dimension estimation for robust detection of ai-generated texts," 2023. [Online]. Available: https://arxiv.org/abs/2306.04723

[5] M. Frohmann, E. V. Epure, G. Meseguer-Brocal, M. Schedl, and R. Hennequin, "Ai-generated song detection via lyrics transcripts," in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, Daejeon, South Korea, 2025, pDF available in user upload.

[6] M. Akinwande, O. Adeliyi, and T. Yussuph, "Decoding ai and human authorship: Nuances revealed through nlp and statistical analysis," *International Journal on Cybernetics & Informatics*, vol. 13, no. 4, pp. 85–103, 2024. [Online]. Available: https://arxiv.org/abs/2408.00769

[7] L. Casini, L. C. Vila, D. Dalmazzo, A.-K. Kaila, and B. L. T. Sturm, "Data-driven analysis of text-conditioned ai-generated music: A case study with suno and udio," 2025. [Online]. Available: https://arxiv.org/abs/2509.11824

[8] I. Ul Haq, M. Pifarré, and E. Fraca, "Novelty evaluation using sentence embedding models in open-ended cocreative problem-solving," *International Journal of Artificial Intelligence in Education*, vol. 34, pp. 1599–1626, 2024. [Online]. Available: https://link.springer.com/article/10.1007/s40593-024-00392-3

[9] S. Shibayama, D. Yin, and K. Matsumoto, "Measuring novelty in science with word embedding," *PLoS ONE*, vol. 16, no. 7, p. e0254034, 2021. [Online]. Available: https://doi.org/10.1371/journal.pone.0254034

[10] MetaBrainz Foundation, "Musicbrainz: The open music encyclopedia," https://musicbrainz.org/, 2025, accessed: 2025-10-2.

[11] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[12] E. Zhou and D. Lee, "Generative artificial intelligence, human creativity, and art," *PNAS Nexus*, vol. 3, 03 2024.

APPENDIX

TABLE II: Comparison of Line Transition Metrics (n=189 AI, n=189 Human)

Metric	Suno	Human	t/F-stat	p-value
Line-to-line cosine distance mean	0.408174	0.390504	-0.8630	0.3887
Line-to-line angle mean	0.640311	0.613182	-0.8443	0.3990
Line-to-line cosine distance variance	0.628506	0.601608	1.0447	0.7646
Line-to-line angle variance	1.547292	1.480928	1.0448	0.7641

TABLE III: GroupKFold Random Forest Classification Accuracy Using Different Feature Sets

Classifier	Final Accuracy
Centroid Only	0.9327
Angles + Distances Only	0.7356
Full Feature Set (centroid + angle/dist)	0.9375

Note: For all classifiers, we used 600 ”trees”, a max depth of 20, a minimum sample split of 4, a leaf node sample minimum of 2, maximum feature count set to the square root of the 768D embeddings, and a random starting state of 42.